16. Exercises - Thermodynamics

1.

a.

i. State what is meant by the internal energy of a system.

.....

.....[2]

ii. Explain why, for an ideal gas, the change in internal energy is directly proportional to the change in thermodynamic temperature of the gas.

.....

.....

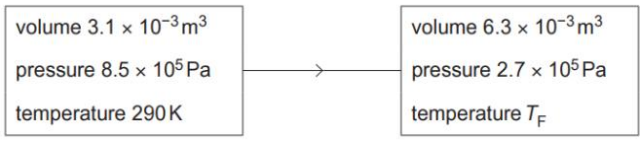
.....[3]

b. A cylinder of volume 1.8×10^4 cm^3 contains helium gas at pressure 6.4×10^6Pa and temperature 25°C. Helium gas may be considered to be an ideal gas consisting of single atoms. Calculate the number of helium atoms in the cylinder.

number = ...[3]

[Total: 8]

2. An ideal gas has a volume of 3.1×10^{-3} m^3 at a pressure of 8.5×10^5 Pa and a temperature of 290 K, as shown in Fig.



The gas suddenly expands to a volume of 6.3×10^{-3} m^3. During the expansion, no thermal energy is transferred. The final pressure of the gas is 2.7×10^5 Pa at temperature T_F, as shown in Fig.

a. Show that the number of gas molecules is 6.6×10^{23}. [3]

b.

i. Calculate the final temperature T_F of the gas.

T_F = ...K [1]

ii. Calculate the increase in internal energy ΔU of the gas.

ΔU = ...J [3]

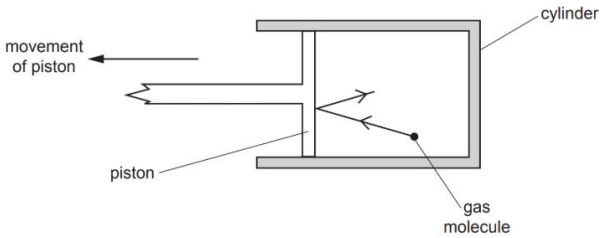
c. Use the first law of thermodynamics to explain why the external work w done on the gas during the expansion is equal to the increase in internal energy in (b)(ii).

.....

.....[2]

[Total: 9]

3. An ideal gas is contained in a cylinder by means of a movable frictionless piston, as illustrated in Fig.



Initially, the gas has a volume of 1.8×10^{-3} m3 at a pressure of 3.3×10^{5} Pa and a temperature of 310 K.

a. Show that the number of gas molecules in the cylinder is 1.4×10^{23}.[2]

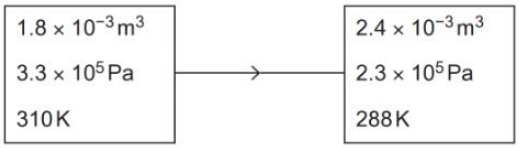
b. Use kinetic theory to explain why, when the piston is moved so that the gas expands, this causes a decrease in the temperature of the gas.

.....

.....

.....[3]

c. The gas expands so that its volume increases to 2.4×10^{-3} m^3 at a pressure of 2.3×10^{5} Pa and a temperature of 288 K, as shown in Fig.



The work done by the gas during the expansion is 76 J.

Calculate and explain whether thermal energy is transferred to or from the gas during the expansion.

.....

.....

.....

.....[4]

[Total: 9]

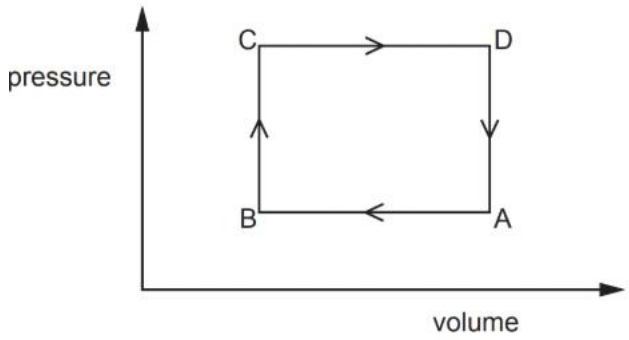
4.

a. State the first law of thermodynamics. Identify the meaning of any symbols that you use.

.....

.....[2]

b. The state of an ideal gas is continuously changed according to the cycle ABCDA shown in Fig.



i. Complete Table for the changes A to B and B to C by placing two ticks (✔) in each row.

change	change in internal energy			work done on gas		
	decrease	no change	increase	negative	zero	positive
A to B						
B to C						

[4]

ii. Use the first law of thermodynamics to describe and explain the energy transfers associated with one complete cycle ABCDA.

.....

.....

.....[3]

[Total: 9]

5. A fixed mass of an ideal gas has a volume V and a pressure p.

The gas undergoes a cycle of changes, X to Y to Z to X, as shown in Fig.

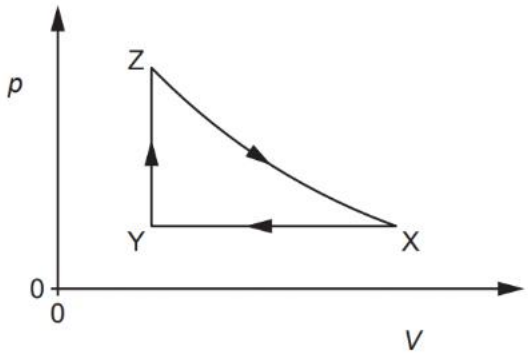


Table shows data for p, V and temperature T for the gas at points X, Y and Z.

	$p/10^5 pa$	$V/10^{-3}$ m^3	T/K
X	1.5	4.2	540
Y			230
Z	5.1		782

a. State the change in internal energy ΔU for one complete cycle, XYZX.

ΔU = ...J [1]

b. Calculate the amount n of gas.

n = ... mol [2]

c. Complete Table

Use the space below for any working. [2]

d.

i. The first law of thermodynamics for a system may be represented by the equation

$$\Delta U = q + W$$

State, with reference to the system, what is meant by:

ΔU: .....

q: .....

W: ..... [3]

ii. Explain how the first law of thermodynamics applies to the change Z to X.

.....

.....[2]

[Total: 10]